

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

$s = (s_0, \dots, s_{15}) \quad A = \{\uparrow, \downarrow, \leftarrow, \rightarrow\}$

$s_i \in \{0, \dots, 15\}$

$\text{Card}(s) = \frac{16!}{2}$

```

class NPuzzle(SearchPb):
    def __init__(self, n, sf=None):
        self.n = n
        if sf != None and len(sf) != n**2:
            raise ValueError

```

```

    def acciones(self, s):
        if not exist(self.a. legales):
            self.a. legales = {}
        for i in range(self.n**2):
            temp = []
            if i // self.n > 0:
                temp.append('↑')
            if i // self.n < self.n-1:
                temp.append('↓')
            if i % self.n > 0:
                temp.append('←')
            if i % self.n < self.n-1:
                temp.append('→')
            self.a. legales[i] = temp[i:]
        return self.a. legales[i:]
    def sucesor(self, s, a):

```